Corporate Finance – Business Valuation (Course 20127)

Dividend Discount Models (DDM)

Milan, October 14th-15th, 2026

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Plan for the day

Last time

- Estimating future Growth of Earnings and Cash-Flows
- Understanding the Relationship between Growth and TV

Today

- Main question for today:
  - learning how to implement a DDM valuation
  - understanding the impact of different Growth Scenarios on a DDM valuation

For Next Time

- Review the material covered in class

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Agenda

1. **Dividend Discount Model (DDM)**
    a. The Gordon Growth Model
    b. The Two-Stage DDM
    c. The Three-Stage DDM
    d. Stock Buybacks and Pay-out Ratio
2. **The Value of Growth**
3. **ABN Example: Two-Stage DDM**
    a. Inputs
    b. Valuation
4. **Takeaways**

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Dividend Discount Models (DDM)

- **DDM is the simplest (and most intuitive) model for valuing equity**
- **It assumes the value of a stock is the PV of its expected dividends:**

$$EqV_t = \sum_{t=1}^{t=\infty} \frac{\mathbb{E}(DPS_t)}{(1+K_e)^t}$$

Where:

EqV_t = Equity Value per share at time t

$E(DPS)_t$ = Expected DPS at time t

K_e = Cost of equity (required return on equity)

- **When an investor buys a stock, s/he expects two types of cashflows:**
  - ✓ Dividends during the holding period
  - ✓ Price at the end of the holding period
- **In this class, we'll study three versions of the DDM:**
  - ✓ The Gordon Growth Model
  - ✓ The Two-Stage DDM
  - ✓ The Three-Stage DDM

Choice of the model depends on growth expectation for the firm

DDM: The Gordon Growth Model

- **<u>Steady state</u> firms with DPS growing at a constant rate forever:**

$$EqV_t = \frac{\mathbb{E}(DPS_{t+1})}{(K_e - g)}$$

Where:

EqV_t	= Equity Value per share at time t
$E(DPS)_t$	= Expected DPS at time t
K_e	= Cost of equity (required return on equity)
g	= Stable growth rate

- **Stable growth rate:**
  - ✓ $g = Retention\ Ratio\ (b) \cdot ROE$
  - ✓ Growth of DPS = growth of EPS (constant b)
- **Limitations:**
  - ✓ Extremely (exponentially) sensitive to assumptions about g
- **Firms model works best for:**
  - ✓ Firms growing at a rate ≤ nominal growth rate in the economy (otherwise they would not be in steady state yet)
  - ✓ Firms with well-established payout policies they intend to continue

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DDM: The Gordon Growth Model

- **Ke can be derived from the Gordon Model:**

$$EqV_t = \frac{\mathbb{E}(DPS_{t+1})}{(K_e - g)} \rightarrow K_e = \frac{\mathbb{E}(DPS_{t+1})}{EqV_t} + g = \frac{DPS_t(1+g)}{Price\ per\ share_t} + g$$

- **How? Rearranging the equation to express K_e in terms of P/E:**

Substituting DPS with EPS and Payout:

$$K_e = \frac{EPS_t \cdot Payout\ (1+g)}{Price\ per\ share_t} + g$$

Dividing the RHS ratio by EPS:

$$K_e = \frac{Payout(1+g)}{P_t/EPS_t} + g$$

- ✓ The denominator (P_t/EPS_t) is the firm's P/E calculated from the profits of the last available period
- ✓ The main difference between DDM and P/E is that DDM is based on the growth of DPS while P/E is based on the growth of EPS
- ✓ If g is estimated based on historical records of earnings or dividends, major differences may arise

DDM: The Gordon Growth Model

- **The Gordon Model can be useful to value a business when:**
  - ✓ The business model and industrial context are such to believe cashflows will grow constantly in the **long-term** (at least 20-30y)
  - ✓ It is better to apply this method to industries growing in line with the whole economy (eg: utilities, banks, …)
  - ✓ Alternatively, it is better to use models based on **limited** growth assumptions

- **The Gordon Model (and each DDM valuation) is based on the following assumptions:**
  - ✓ The risk-adjusted discount rate is constant in the long-term
  - ✓ The growth rate is constant in the long-term
  - ✓ The payout ratio is constant in the long-term

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DDM: The Gordon Growth Model (example)

- **During LFY a firm reported EPS of $3.47 and DPS of $2.22**
- **Assume that:**
  - ✓ RF is 3.5%
  - ✓ β is 0.8x (industry average)
  - ✓ ERP is 5%
  - ✓ g = RF
  - ✓ ROE is 9.8%
- **What is the Cost of Equity?**

$$K_e = R_f + \beta \cdot ERP = .035 + .8 \cdot .05 = 7.5\%$$

- **What is the EqV per share?**

$$EqV_t = \frac{\mathbb{E}(DPS_{t+1})}{(K_e - g)} = \frac{2.22 \cdot (1.035)}{(.075 - .035)} = \$57.44$$

- **What is the Retention Rate?**

$$RR = 1 - \frac{DPS}{EPS} = 1 - \frac{2.22}{3.47} = 36.0\%$$

- **Does g make sense relative to firm's fundamentals?**

$$E(g) = RR \cdot ROE = .36 \cdot .098 = 3.5\%$$

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DDM: The Two-Stage DDM

- **Two-Stage DDM allows for two periods of growth: (i) an initial phase where g is not stable (ii) a subsequent steady state phase:**

$$EqV_0 =$$

$$= \sum_{t=1}^{t=n} \frac{\mathbb{E}(DPS_t)}{(1+K_{e,hg})^t} + \frac{\mathbb{E}(P_n)}{(1+K_{e,hg})^n} = \sum_{t=1}^{t=n} \frac{\mathbb{E}(DPS_t)}{(1+K_{e,hg})^t} + \frac{\mathbb{E}(DPS_{n+1})}{(K_{e,st}-g_n)(1+K_{e,hg})^n}$$

Where:

$E(DPS)_t$	= Expected DPS in t
K_e	= Cost of equity (hg = high growth; st = stable growth)
$E(P)_n$	= Expected stock price at the end of n (ie: Terminal Value)
g	= Growth rate for the first n periods (ie: first stage)
g_n	= Stable growth rate

- **If g and payout in the first stage are constant:**

$$EqV_0 = \frac{DPS_0(1+g)\left[1-\frac{(1+g)^n}{(1+K_{e,hg})^n}\right]}{(K_{e,hg}-g)} + \frac{\mathbb{E}(DPS_{n+1})}{(K_{e,st}-g_n)(1+K_{e,hg})^n}$$

Growing Annuity

DDM: The Two-Stage DDM

- **Stable growth rate:**
  - ✓ $g = RR \cdot ROE = (1 - Payout) \cdot ROE$
  - ✓ Payout should be consistent with g → stable firms have higher payouts, all else equal
  - ✓ Long-run ROE should be consistent with industry perspectives
- **Cost of equity:**
  - ✓ As we have different growth phases, we should adjust Ke in the different periods (adjustments may be related to β and capital structure)
- **Limitations:**
  - ✓ It is difficult to estimate the length of the first growth stage
  - ✓ The shift from the first growth rate to stable growth is sudden (it should be progressive)
  - ✓ Focus on dividends can lead to skewed estimates of value for non-dividend-paying firms (or firms not paying out what they could)
- **Firms model works best for:**
  - ✓ Firms having a finite period competitive advantage (barriers to entry, patents, …)
  - ✓ Firms with modest growth rate in the initial phase
  - ✓ Firms committed/able to maintain a payout policy of residual cash

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DDM: The Three-Stage DDM

- **Three-Stage DDM allows for three periods of growth: (i) initial phase of high g, (ii) period of declining g (iii) a final stable g:**

$$EqV_0 = \sum_{t=1}^{t=n_1} \frac{EPS_0 \cdot (1+g_a)^t \cdot \Pi_a}{\left(1+K_{e,hg}\right)^t} + \sum_{t=(n_1+1)}^{t=n_2} \frac{\mathbb{E}(DPS_t)}{\left(1+K_{e,tr}\right)^t} + \frac{\mathbb{E}\left(EPS_{n_2}\right) \cdot (1+g_n) \cdot \Pi_n}{\left(K_{e,st}-g_n\right) \cdot \left(1+K_{e,tr}\right)^{n_2}}$$

Terminal Value

Where:

EPS_t = EPS in t

DPS_t = DPS in t

k_e = Cost of equity (hg = high growth; tr = transition; st = stable growth)

g_a = Growth rate for the first n_1 periods (ie: first stage)

g_n = Stable growth rate

Π_a = Payout in high growth period

Π_n = Payout in stable growth period

n_1 = # of periods in high growth stage

$n_2 - n_1$ = # of periods in transition stage

DDM: The Three-Stage DDM

- **Stable growth rate:**
  - ✓ $g = RR \cdot ROE = (1 - Payout\ ) \cdot ROE$
  - ✓ Payout should be consistent with g → stable firms have higher payouts, all else equal
  - ✓ Long-run ROE should be consistent with industry perspectives
- **Cost of equity:**
  - ✓ As we have different growth phases, we should adjust Ke in the different periods (adjustments may be related to β and capital structure)
- **Limitations:**
  - ✓ It requires more inputs to be implemented
  - ✓ Focus on dividends can lead to skewed estimates of value for non-dividend-paying firms (or firms not paying out what they could)
- **Firms model works best for:**
  - ✓ Firms changing growth and other dimensions (payout, risk, …) over time
  - ✓ Firms having a competitive advantage which, after a period, is expected to decline gradually
  - ✓ Firms committed/able to maintain a payout policy of residual cash

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DDM: The Three-Stage DDM (example)

- **During LFY a firm reported EPS of $3.56 and DPS of $2.26. Ke is 8.45%**
- **Assume the firm will reinvest 36.5% of its earnings generating ROE = 25%**
- **What is the expected growth rate as of LFY?**

$$g = RR \cdot ROE = .25 \cdot .365 = 9.10\%$$

- **Expected dividends over the next 5 years are shown below**

Year	*1*	*2*	*3*	*4*	*5*
EPS	3.88	4.23	4.61	5.03	5.49
Payout	1-.364	1-.364	1-.364	1-.364	1-.364
DPS	2.47	2.69	2.93	3.20	3.49
PV of DPS	2.27	2.29	2.30	2.31	2.33

- **After year 5, we allow for a transition of 5 years to stable g after year 10.**
- **In the stable growth phase, assume(1):**
  - ✓ Expected g = 3%
  - ✓ ROE = 15%
  - ✓ Ke = 9%
- **What is the expected payout in the stable phase?**

$$Payout = 1 - g/ROE = 1 - .03/.15 = 80\%$$

(1) Each figure will be reached linearly over time

DDM: The Three-Stage DDM (example)

- **What is the TV per share at the end of year 10?**

Year	6	7	8	9	10
Expected g	7.88%	6.66%	5.44%	4.22%	3.00%
EPS	5.93	6.32	6.66	6.95	7.15
Payout	66.88%	70.16%	73.44%	76.72%	80.00%
DPS	3.96	4.43	4.89	5.33	5.72
Ke	8.56%	8.67%	8.78%	8.89%	9.00%
Cumulative Ke(1)	1.6286	1.7698	1.9252	2.0964	2.2850
PV of DPS	2.43	2.51	2.54	2.54	2.50

- **At the end of year 10, the (undiscounted) TV is:**

$$TV = \frac{\mathbb{E}(DPS_{11})}{(Ke_{stable} - g_{stable})} = \frac{7.15 \cdot (1.03) \cdot (.80)}{(.09 - .03)} = \$98.24$$

- **What is the expected value per share based on these forecasts?**

$$EqV = PV(DPS) + PV(TV) = \$24.03 + \frac{\$98.24}{2.2850} = \$67.03$$

(1) Changing Ke requires a cumulated discount rate. For instance, the cumulated Ke for year 7 is: $Ke_7 = (1.0845)^5 \cdot (1.0856) \cdot (1.0867)$

The Value of Growth from a Two-Stage DDM

- **In any model, it is possible to extract the portion of value that can be attributed to growth and to break it down into the portion attributable to high growth and to stable growth**
- In the case of the 2-stage DDM, this can be accomplished as follows:

$$EvQ = \left[\sum_{t=1}^{t=n} \frac{\mathbb{E}(DPS_t)}{(1+k_{e,hg})^t} + \frac{\mathbb{E}(DPS_{n+1})}{(k_{e,st}-g_n)(1+k_{e,hg})^n} - \frac{\mathbb{E}(DPS_1)}{(k_{e,st}-g_n)}\right] + \text{Value of High Growth}$$

$$+ \left(\frac{\mathbb{E}(DPS_1)}{(k_{e,st}-g_n)} - \frac{DPS_0}{k_{e,st}}\right) + \text{Value of Stable Growth}$$

$$+ \frac{DPS_0}{k_{e,st}} \quad \text{Value of Assets in Place}$$

- **Value of high g =** Total value – Value of firm as a stable growth firm
- **Value of stable g =** Value of firm as a stable growth firm – Value of firm with no growth
- **Value of assets in place =** Value of firm with *no growth* (DPS = EPS)

Two-Stage DDM: ABN - Inputs

- **Market inputs:**
  - ✓ Long-term Rf (in Euros) = 5.02%
  - ✓ Risk premium = 4% (US premium: Netherlands is AAA rated)
- **Current EPS = € 1.60**
- **Current DPS = € 0.60**

Variable	*High Growth Phase*	*Stable Growth Phase*
Length	5 years	Forever after year 5
ROE	15.56%	15% (Industry average)
Payout	37.5%	66.67%
Retention	62.5%	33.33% (b=g/ROE)
Expected *g*	.1556*.625=.0973	5%
B	0.95	1.00
Ke	5.02%+0.95(4%)=8.82%	5.02%+1.00(4%)=9.02%

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Two-Stage DDM: ABN - Valuation

Year	EPS	DPS	PV of DPS
1	1.76	0.66	0.65
2	1.93	0.72	0.61
3	2.11	0.79	0.62
4	2.32	0.87	0.62
5	2.54	0.95	0.63

- **Expected EPS in year 6 =** $2.54 \cdot (1.05) =$ **€ 2.67**
- **Expected DPS in year 6 =** $2.67 \cdot 0.67 =$ **€ 1.78**
- **TV (in year 5) =** $1.78/(.09 - .05) =$ **€ 44.31**
- **PV of TV =** $44.31/(1.09)^5 =$ **€ 29.04**

EqV = 0.61+0.61+0.62+0.62+0.63+29.04 = € 32.12

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DDM: Buybacks and Payout

- **US firms have increasingly turned to buybacks as a way of returning cash to shareholders (also for taxation of capital gain < dividends)**
  - ✓ Focusing only on DPS may miss an important portion of cash returned to shareholders through repurchases
  - ✓ We can estimate an **augmented payout ratio**:

$$Augmented\ Payout = \frac{(Dividends\ + Buybacks)}{Net\ Income}$$

- **Buybacks could be smoothed (averaged) over time**
  - ✓ If buybacks have been financed by debt to increase the leverage, the **modified augmented payout** is:

$$Modified\ Augmented\ Payout\ = \frac{(Dividends\ + Buybacks - Debt\ Issues)}{Net\ Income}$$

$$Augmented\ Growth\ Rate = (1 - Augmented\ Payout\ ) \cdot ROE$$

DDM: Buybacks and Payout

- **On Jan 1st, the S&P is trading at 1,257.64, and DPS on the index from last year were $23.12:**
  - ✓ The expected g in both EPS and DPS for the next 5 years is 6.95%
  - ✓ After year 5, both EPS and DPS are expected to grow at 3.29%
  - ✓ The cost of capital is 8.5%
- **What are the projected DPS and value per share?**

Year	*1*	*2*	*3*	*4*	*5*
$\mathbb{E}(DPS)$	24.73	26.45	28.28	30.25	32.35

$$EqV = \frac{24.73}{(1.085)} + \frac{26.45}{(1.085)^2} + \frac{28.28}{(1.085)^3} + \frac{30.25}{(1.085)^4} + \frac{32.35}{(1.085)^5} + \frac{32.35 \cdot (1.0329)}{(.085 - .0329) \cdot (1.085)^5} = 537.28$$

- **Was the index fairly priced @ Jan 1st?**
  - ✓ The index looks overvalued
  - ✓ This valuation does <u>not</u> consider buybacks though

DDM: Buybacks and Payout

- **Buybacks from last year were $30,84**
- **Adding buybacks to dividends we arrive at a cash remuneration of $53.96 per share ($23,12 + $30,84)**
- **Applying the same parameters as before using augmented dividends we arrive at:**

$$EqV = \frac{57.71}{(1.085)} + \frac{61.72}{(1.085)^2} + \frac{66.01}{(1.085)^3} + \frac{70.60}{(1.085)^4} + \frac{75.51}{(1.085)^5} + \frac{75.51 \cdot (1.0329)}{(.085 - .0329) \cdot (1.085)^5} = 1{,}253.97$$

- **With buybacks incorporated the index looks (almost) fairly priced**

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Takeaways

- **DDM is widely used to value Financial Institution Groups (FIG), especially banks since they pay predictable/constant DPS**
- **DDM is very sensitive to assumptions, particularly about the perpetuity *g* rate**
- **To smooth out the impact of *g* on value, one can use models allowing for different growth rates**
- **Consistency is key:**
  - ✓ Payouts should be consistent with *g* rates: stable firms have higher payout ratio, all else equal
  - ✓ Long-run ROE should be consistent with the industry perspective

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